

Otherwise, knowledge about friendship is incomplete in that it may be known that e.g. a is a friend of b , and it may be not known whether b is a friend of a . Finally, there is general knowledge about smoking, friendship, and cancer, namely that smoking causes cancer, friendship is normally symmetric and anti-reflexive, everyone has a friend, and smoking propagates (actively or passively) among friends. All this knowledge is represented in the axioms further below.

Domains:

people, to denote the individuals.

Constants:

$a, b, \dots, h, i, j, \dots, n$, the 14 individuals. Our goal is to learn an adequate embedding for each constant.

$D(a) = D(b) = \dots = D(n) = \text{people}$.

Variables:

x, y ranging over the individuals.

$D(x) = D(y) = \text{people}$.

Predicates:

$S(x)$ for *smokes*, $F(x, y)$ for *friends*, $C(x)$ for *cancer*.

$D(S) = D(C) = \text{people}$. $D(F) = \text{people, people}$.

Axioms:

Let $\mathcal{X}_1 = \{a, b, \dots, h\}$ and $\mathcal{X}_2 = \{i, j, \dots, n\}$ be the two groups of individuals.

Let $\mathcal{S} = \{a, e, f, g, j, n\}$ be the smokers; knowledge is complete in both groups.

Let $\mathcal{C} = \{a, e\}$ be the individuals with cancer; knowledge is complete in $\mathcal{X}_1$ only.

Let $\mathcal{F} = \{(a, b), (a, e), (a, f), (a, g), (b, c), (c, d), (e, f), (g, h), (i, j), (j, m), (k, l), (m, n)\}$ be the set of friendship relations; knowledge is complete if assuming symmetry.

These facts are illustrated in Figure 20a.

We have the following axioms:

$$F(u, v) \quad \text{for } (u, v) \in \mathcal{F} \quad (46)$$

$$\neg F(u, v) \quad \text{for } (u, v) \notin \mathcal{F}, u > v \quad (47)$$

$$S(u) \quad \text{for } u \in \mathcal{S} \quad (48)$$

$$\neg S(u) \quad \text{for } u \in (\mathcal{X}_1 \cup \mathcal{X}_2) \setminus \mathcal{S} \quad (49)$$

$$C(u) \quad \text{for } u \in \mathcal{C} \quad (50)$$

$$\neg C(u) \quad \text{for } u \in \mathcal{X}_1 \setminus \mathcal{C} \quad (51)$$

$$\forall x \neg F(x, x) \quad (52)$$

$$\forall x, y (F(x, y) \rightarrow F(y, x)) \quad (53)$$

$$\forall x \exists y F(x, y) \quad (54)$$

$$\forall x, y ((F(x, y) \wedge S(x)) \rightarrow S(y)) \quad (55)$$

$$\forall x (S(x) \rightarrow C(x)) \quad (56)$$

$$\forall x (\neg C(x) \rightarrow \neg S(x)) \quad (57)$$

Notice that the knowledge base is not satisfiable in the strict logical sense of the word. For instance, f is said to smoke but not to have cancer, which is inconsistent with the rule

$\forall x (S(x) \rightarrow C(x))$. Hence, it is important to adopt a fuzzy approach as done with MLN or a many-valued fuzzy logic interpretation as done with LTN.

Grounding:

$\mathcal{G}(\text{people}) = \mathbb{R}^5$. The model is expected to learn embeddings in $\mathbb{R}^5$.

$\mathcal{G}(a \mid \theta) = \mathbf{v}_\theta(a), \dots, \mathcal{G}(n \mid \theta) = \mathbf{v}_\theta(n)$. Every individual is associated with a vector of 5 real numbers. The embedding is initialized randomly uniformly.

$\mathcal{G}(x \mid \theta) = \mathcal{G}(y \mid \theta) = \langle \mathbf{v}_\theta(a), \dots, \mathbf{v}_\theta(n) \rangle$.

$\mathcal{G}(S \mid \theta) : x \mapsto \text{sigmoid}(\text{MLP_S}_\theta(x))$, where MLP_S_θ has 1 output neuron.

$\mathcal{G}(F \mid \theta) : x, y \mapsto \text{sigmoid}(\text{MLP_F}_\theta(x, y))$, where MLP_F_θ has 1 output neuron.

$\mathcal{G}(C \mid \theta) : x \mapsto \text{sigmoid}(\text{MLP_C}_\theta(x))$, where MLP_C_θ has 1 output neuron.

The MLP models for S, F, C are kept simple, so that most of the learning is focused on the embedding.

Learning:

We use the stable real product configuration to approximate the operators. For $\forall$, we use A_{pME} with $p = 2$ for all the rules, except for rules (52) and (53), where we use $p = 6$. The intuition behind this choice of p is that no outliers are to be accepted for the friendship relation since it is expected to be symmetric and anti-reflexive, but outliers are accepted for the other rules. For $\exists$, we use A_{pM} with $p = 1$ during the first 200 epochs of training, and $p = 6$ thereafter, with the same motivation as that of the schedule used in Section 4.4. The formula aggregator is approximated by A_{pME} with $p = 2$.

Figure 19 shows the satisfiability over 1000 epochs of training. At the end of one of these runs, we query $S(x), F(x, y), C(x)$ for each individual; the results are shown in Figure 20b. We also plot the principal components of the learned embeddings [51] in Figure 21. The friendship relations are learned as expected: (56) "smoking implies cancer" is inferred for group 2 even though such information was not present in the knowledge base. For group 1, the given facts for smoking and cancer for the individuals f and g are slightly altered, as these were inconsistent with the rules. (the rule for smoking propagating via friendship (55) is incompatible with many of the given facts). Increasing the satisfaction of this rule would require decreasing the overall satisfaction of the knowledge base, which explains why it is partly ignored by LTN during training. Finally, it is interesting to note that the principal components for the learned embeddings seem to be linearly separable for the smoking and cancer classifiers (c.f. Figure 21, top right and bottom right plots).

Querying:

To illustrate querying in LTN, we query over time two formulas that are not present in the knowledge-base:

$$\phi_1 : \quad \quad \quad \forall p : C(p) \rightarrow S(p) \quad (58)$$

$$\phi_2 : \quad \quad \quad \forall p, q : (C(p) \vee C(q)) \rightarrow F(p, q) \quad (59)$$

We use $p = 5$ when approximating $\forall$ since the impact of an outlier at querying time should be seen as more important than at learning time. It can be seen that as the grounding approaches satisfiability of the knowledge-base, ϕ_1 approaches *true*, whereas ϕ_2 approaches *false* (c.f. Figure 20a).

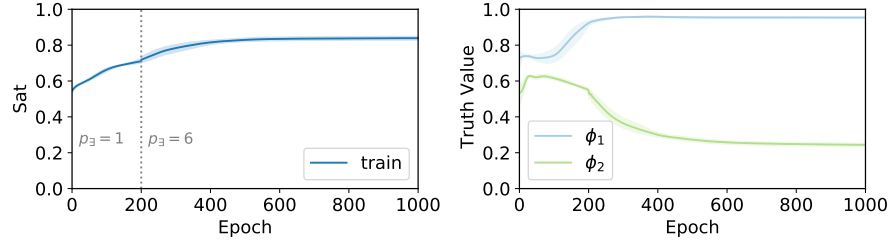
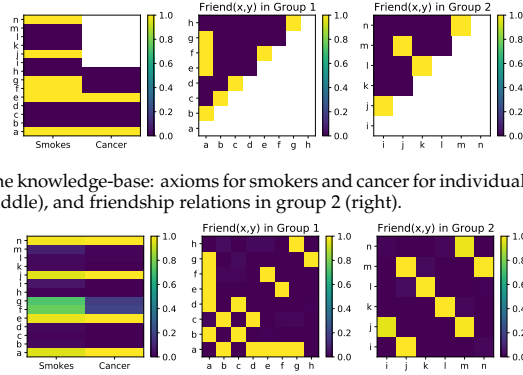


Figure 19: Smoker-Friends-Cancer example: Satisfiability levels during training (left) and truth-values of queries ϕ_1 and ϕ_2 over time (right).



(a) Incomplete facts in the knowledge-base: axioms for smokers and cancer for individuals a to n (left), friendship relations in group 1 (middle), and friendship relations in group 2 (right).

(b) Querying all the truth-values using LTN after training: smokers and cancer (left), friendship relations (middle and right).

Figure 20: Smoker-Friends-Cancer example: Illustration of the facts before and after training.